# DANG THANH TUAN

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## SUMMARY

Artificial Intelligence student at FPT University (GPA 3.6/4.0) with hands-on experience building RAG systems, LLM chatbots and AI agents, from prototype to deployment on AWS. Author of a GraphRAG paper accepted at EAI FISAT 2026 and a paper presented at AICCC 2024. Seeking an AI Engineer role to build RAG, agent and automation features that solve real business problems.

## EDUCATION

**FPT University** — Ho Chi Minh City, Vietnam
Bachelor of Science in Artificial Intelligence; GPA: 3.6/4.0 — 2023 – 2027 (expected)

## PROJECT EXPERIENCE

**SlotHub – AI-Powered Learning Platform** — GitHub
*AI Engineer*, AWS First Cloud AI Journey Bootcamp — Jan 2026 – May 2026

- Built, with a team in the AWS Study Group bootcamp, an education platform for teachers and students featuring an LLM chatbot, AI exercise generation and personalized learning paths.
- Built the AI chatbot agent on Amazon Bedrock AgentCore that answers student and teacher questions over course content.
- Automated exercise generation by difficulty level through prompt design, replacing manual question writing.
- Generated personalized learning paths from student performance data.
- Exposed the AI modules as FastAPI REST services, integrated them with the backend, frontend and PostgreSQL, and deployed them with Docker on AWS (EC2, S3, RDS).

*Technologies:* Python, Amazon Bedrock AgentCore, LangChain, Hugging Face, FastAPI, PostgreSQL, Docker, AWS

**MD-GraphRAG – Markdown-Aware GraphRAG for Long Documents** — GitHub
*AI Research Engineer*, Research Project (accepted at EAI FISAT 2026) — Feb 2026 – Jun 2026

- Designed an end-to-end RAG architecture covering ingestion, chunking, indexing, retrieval and generation for long-document question answering.
- Built Markdown-aware chunking that preserves headings and sections, improving the context passed to the LLM.
- Implemented two-channel retrieval combining vector search (ChromaDB, pgvector, BGE-M3) with knowledge-graph traversal.
- Extracted knowledge-graph triplets with open-source LLMs (Llama 3.1 8B, Qwen 2.5 7B) served locally via Ollama.
- Benchmarked 4 pipeline variants (chunking × knowledge-graph ablation) on 7 QA datasets, including HotpotQA, QASPER and PubMedQA, scored with RAGAS.

*Technologies:* Python, LangChain, ChromaDB, pgvector, NetworkX, Ollama, Llama 3.1, Qwen 2.5, BGE-M3, RAGAS

## AWARDS, COMPETITIONS & PUBLICATIONS

**Paper accepted**, "MD-GraphRAG: Synergizing Markdown-Aware Chunking and Knowledge Graph Enrichment for Retrieval-Augmented Generation", EAI FISAT (Q4 journal) — 2026
**1st Prize**, Jetson AI Racer Challenge (autonomous racing car on NVIDIA Jetson) — 2026
**3rd Prize**, Datathon "The Gridbreaker: Breaking Business Boundaries", VinUni Data Science & AI Club — 2026
**Honorable Mention**, SEAL Hackathon Spring, FPT University — 2026
**Published and presented paper** at the AICCC Conference, Tokyo, Japan (ACM Digital Library) — 2024
**National Consolation Prize**, Vietnam Young Informatics Contest; **3 provincial First Prizes** in IT & science contests, Gia Lai — 2022 – 2023

## CORE COMPETENCIES

**RAG & Retrieval:** design RAG pipelines end to end, from structure-aware chunking and embeddings to hybrid vector + knowledge-graph retrieval (LangChain, ChromaDB, pgvector).
**LLM Apps & Agents:** build chatbots and AI agents with prompt engineering, managed agent platforms (Amazon Bedrock AgentCore) and self-hosted open-source LLMs (Ollama, Llama, Qwen).
**Evaluation:** measure answer quality with RAGAS and ablation studies to choose the best pipeline based on evidence.
**Deployment:** package AI features as FastAPI REST services with PostgreSQL, Docker and AWS (EC2, S3, RDS).

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, C++, R
**AI / ML Frameworks:** LangChain, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, OpenCV
**LLM & RAG:** Amazon Bedrock AgentCore, Ollama, Llama, Qwen, BGE-M3, ChromaDB, pgvector, NetworkX, RAGAS
**Backend, Cloud & Tools:** FastAPI, PostgreSQL, AWS (EC2, S3, RDS), Docker, Git, GitHub
**English:** working proficiency (IELTS 6.0 equivalent); wrote and presented research papers in English.